

Bitcoin intraday market microstructure research

Preliminary search for uncorrelated alpha using 1-minute BTC data, rare-event prediction, walk-forward validation and risk-controlled trading rules.

974k

1-minute bars
BTC-USD OHLC

50

features
microstructure proxy

30-60m

forecast horizons
big-move labels

50 bps

event threshold
up/down targets

Core research hypothesis

The project does not try to forecast the exact Bitcoin price. It searches for rare intraday states in which the probability of a large short-term move becomes higher than the unconditional base rate. The research objective is to develop a rule-based, risk-controlled trading engine that can potentially generate alpha less dependent on passive BTC exposure.

Professional interpretation

The current evidence is promising but preliminary. Classification quality is strong relative to rare-event base rates, especially for Random Forest and MLP probability signals, but trading performance is not yet stable across all validation designs. The correct positioning is: research toward decorrelated alpha, not a production-ready strategy.

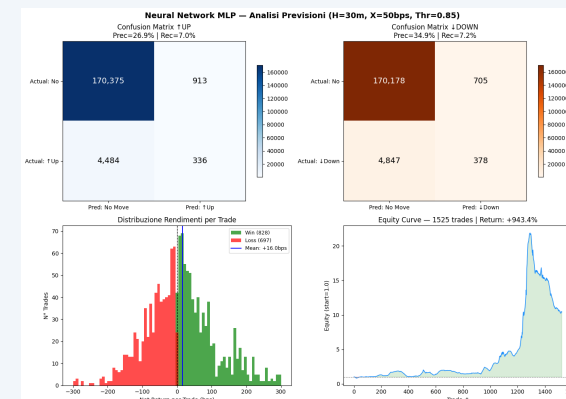


Exhibit: MLP rare-event classification and trade diagnostics

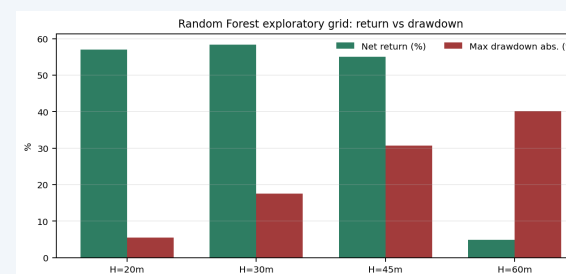
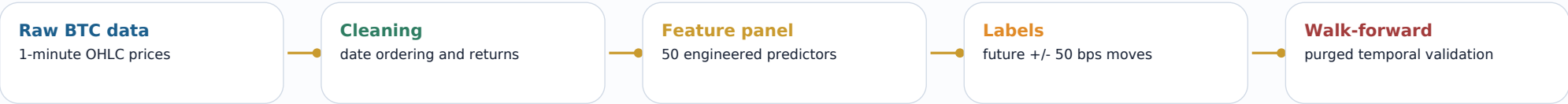


Exhibit: RF exploratory return/drawdown grid

1. Data architecture: from raw BTC bars to a no-leak feature panel



Dataset construction logic

The raw dataset contains 974,098 one-minute observations, covering the period from 2021-12-31 04:03 to 2022-12-14 00:00. The notebook creates a structured file named BTC_RF_features_noleak_1m.csv, where each row represents an investable minute and each predictor is computed using information available before the forecast horizon. This is essential because intraday models are highly exposed to look-ahead bias.

Microstructure-inspired feature set

Feature family	Examples	Market information captured
Return memory	ret_1m, logret_1...240	recent price pressure and short-term autocorrelation
Candle structure	range_pct, body_pct, wick ratios, close_pos	intraday imbalance, rejection, pressure
Momentum	mom_mean/sum 5-120	directional acceleration across horizons
MA distance	dist_ma_15...240	trend location and distance from local fair value
Volatility regime	vol_15...240, vol_ratio	realized volatility compression/expansion
Local extremes	dist_hh/ll, squeeze_30_120	proximity to breakout/breakdown zones

Why this matters

- Minute data allows the model to focus on temporary dislocations rather than long-term BTC beta.
- Features are derived from price dynamics, volatility, candle shape and local extremes, not from fundamental data.
- The target is rare by construction, so precision lift matters more than raw accuracy.
- Embargo and walk-forward splits reduce train-test contamination.

Primary target	Big move +/- 50 bps
Date range	2021-12 to 2022-12
Feature count	50
Columns	60
Rows loaded	974,098

2. Predictive engine: rare-event classification, not price forecasting

Neural Network MLP

Deep nonlinear classifier using standardized feature input, dropout, validation split and walk-forward retraining. Best used as a probability signal; high turnover version is diagnostic, not deployable.

Random Forest

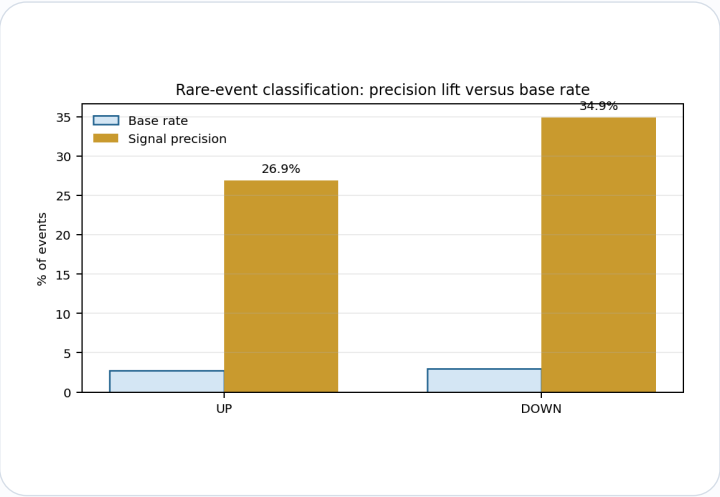
Tree ensemble classifier that captures nonlinear thresholds in volatility, momentum and local range variables. It showed the strongest exploratory trade-level results, but with configuration sensitivity.

Risk controls

The backtest includes round-trip costs, probability thresholds, cooldown, max trades/day, one-position logic and TP/SL exits. These controls convert prediction into an executable trading process.

MLP classifier diagnostic - H=30m, X=50bps, threshold=0.85

Signal	Base rate	Signals	Precision	Recall	TP	FP	Precision lift
UP >= 50 bps in 30m	2.74%	1,249	26.9%	7.0%	336	913	9.8x
DOWN <= -50 bps in 30m	2.97%	1,083	34.9%	7.2%	378	705	11.8x



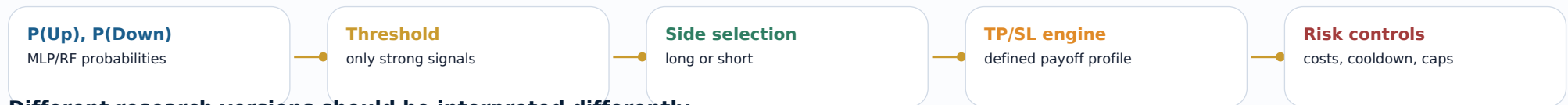
How to read the signal

The base rate of large moves is very low: only 2.74% of OOS bars trigger an UP event and 2.97% trigger a DOWN event. The MLP emits far fewer signals than bars and achieves precision of 26.9% for UP and 34.9% for DOWN. This is not high in absolute terms, but it is materially above the unconditional probability of the event. The model is therefore better interpreted as a filter that concentrates rare opportunities, not as a complete standalone trading system.

Key point

For rare-event trading, recall can be low if precision and payoff asymmetry are sufficient. The goal is to select only a small subset of high-conviction states.

3. Trading architecture: turning probability scores into controlled exposure



Different research versions should be interpreted differently

Research run	Trades	Hit rate	Mean net/trade	Net return	Professional reading
MLP H30 diagnostic	1,525	54.3%	+16.0 bps	+943.4%	High turnover / event concentration
MLP H60 professional	60	35.0%	-6.0 bps	-3.64%	Low-trade, stricter execution
RF exploratory grid	94-148	50.0%-67.0%	+31.8 to +47.9 bps	+55%-58%	Promising but short validation window

Research pipeline, not a sales pitch

The high-trade MLP run proves that the model can locate periods with strong intraday directional opportunity, but the result is concentrated and heavily exposed to regime/event risk. The stricter 60-minute professional version is more realistic, but it loses money, showing that probability quality alone is not enough. The Random Forest runs show stronger preliminary edge, but still require longer validation before being marketed as stable alpha.

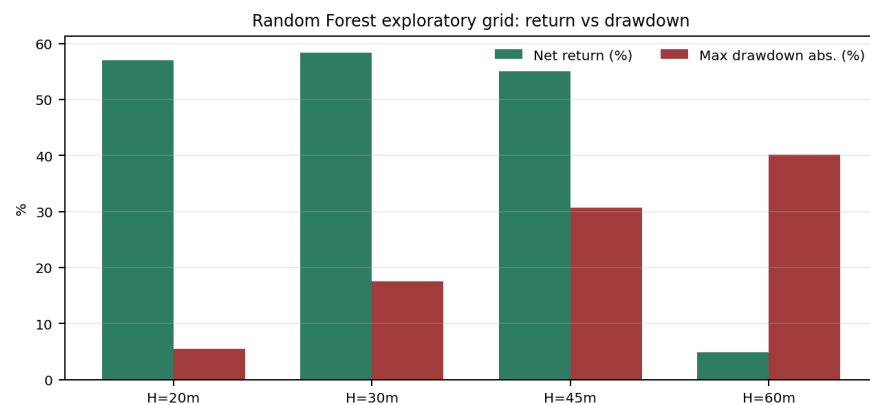
Risk-management layer

- Round-trip trading cost: 8 bps in the professional MLP run.
- TP/SL grids: 80/40, 100/50 and 120/60 bps tested.
- Cooldown: 60-120 minutes to reduce overtrading after exits.
- Max trades/day: 3-5 in the low-trade setup.
- One position at a time to avoid overlapping exposure.

4. Empirical evidence: Random Forest is promising, but robustness is not solved

Random Forest exploratory grid results

Model	H	X	Net return	Max DD	Mean/trade	Hit rate	Trades	AUC Up/Down
RF	20m	50bps	+57.07%	-5.44%	+46.5 bps	75.5%	98	0.778/0.789
RF	30m	50bps	+58.38%	-17.48%	+31.8 bps	50.0%	148	0.748/0.749
RF	45m	50bps	+55.09%	-30.67%	+47.9 bps	67.0%	94	0.724/0.720
RF	60m	50bps	+4.86%	-40.22%	+6.9 bps	60.5%	86	0.694/0.698



Interpretation of results

The RF grid shows that the model can identify high-probability microstructure states with positive average trade return. However, results are sensitive to horizon and holding time. H=20m and H=30m deliver similar net return with materially different drawdown; H=45m keeps high mean return but increases drawdown; H=60m loses most of the edge. This suggests that the signal is short-lived and should be treated as an intraday microstructure effect rather than a broad directional BTC forecast.

Annualized metrics are intentionally not emphasized: the test window is short and intraday strategies can produce misleading annualized ratios.

5. Microstructure reading: the edge appears event-driven and regime-dependent

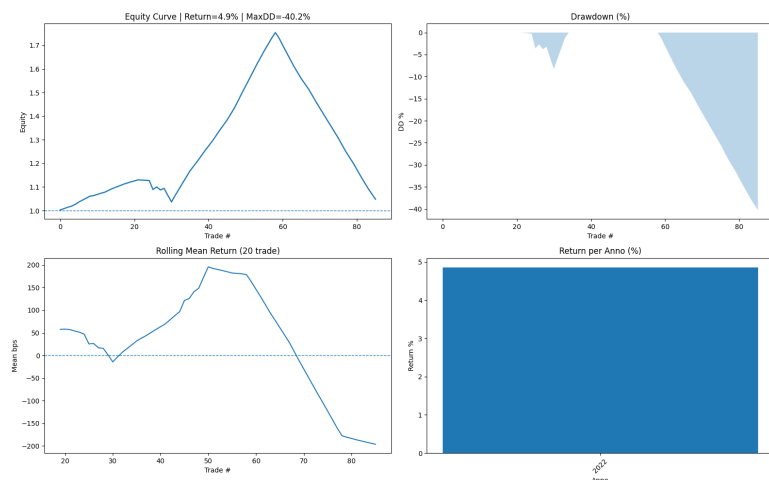


Exhibit: robustness check from trade log - equity, drawdown, rolling mean and annual return

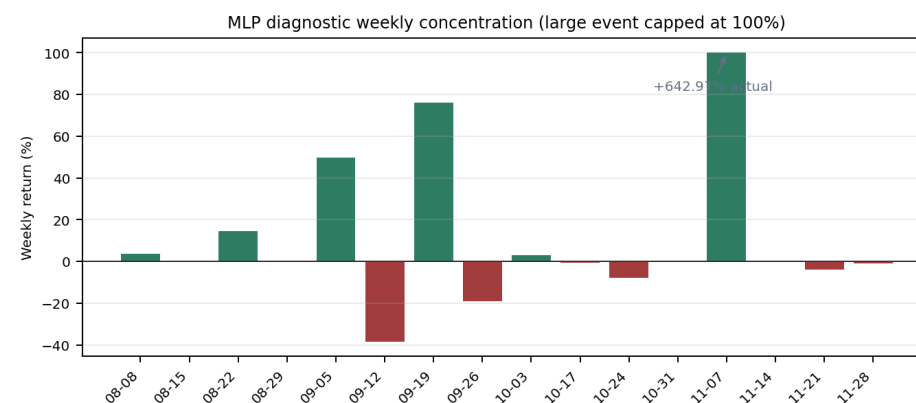


Exhibit: diagnostic weekly concentration - regime dependence and event risk

What the strategy is actually detecting

The feature set is microstructure-inspired: short returns, candle shape, wick pressure, range compression, volatility ratios, moving-average distance and distance from recent highs/lows. These variables attempt to detect when the order-flow environment is no longer neutral, even without direct order-book data.

The most important limitation is stability. The model can identify profitable clusters, but those clusters are not evenly distributed. This is consistent with intraday crypto markets, where liquidity, news shocks and forced deleveraging can temporarily dominate normal price formation.

Market pressure

Volatility expansion

Local breakout states

Regime dependence

6. Final assessment: credible research toward decorrelated alpha

The notebook demonstrates a real quantitative research process: data engineering, no-leak feature construction, supervised learning, walk-forward testing, trading rules and risk diagnostics.

What works

Rare-event classifiers show meaningful lift versus the base probability of large intraday moves. Random Forest reaches AUC values around 0.72-0.79 across configurations, and selected RF runs generate positive net exploratory results.

What is not solved

The alpha is not yet stable enough to be presented as deployable. Some performance is concentrated in specific weeks, model configurations produce different outcomes, and the professional low-trade MLP version is negative after costs.

Why it is valuable

The project demonstrates the ability to build a full intraday ML trading research pipeline and to interpret results critically rather than overfitting a backtest. This is highly relevant for quant research and risk-managed portfolio construction.

Research roadmap to make the alpha claim stronger

Next development	Purpose	Professional rationale
Longer sample and multi-exchange data	Test 2023-2026 and exchange robustness	reduce sample and venue-specific bias
BTC benchmark and correlation analysis	Measure beta to BTC buy-and-hold	prove decorrelation instead of assuming it
Real execution model	Slippage, bid-ask, latency, fees	convert paper edge into tradable edge
Regime filter	Trade only in suitable volatility/liquidity states	reduce drawdown and unstable weeks
Position sizing	Volatility targeting and drawdown limits	stabilize P&L path and capital usage
Feature upgrade	Order book, volume, funding, liquidations	move from proxy microstructure to actual microstructure

Portfolio takeaway

This is best presented as a preliminary, risk-aware intraday research framework for finding market-independent alpha - not as final strategy. Its value lies in the full pipeline and the honest robustness analysis.